
Predicting Customer Churn in the Telecommunications Industry using Machine Learning Techniques

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Abstract: Customer churn continues to pose a major threat to revenue stability and long-term competitiveness in the telecommunications industry, particularly within business-to-business markets where customers manage multiple subscriptions and switching costs remain relatively low. As acquisition expenses increase, the ability to identify customers at risk of leaving has become a critical requirement for effective customer relationship management. This study develops and evaluates a machine learning based framework for predicting customer churn using real operational data from a leading Bulgarian telecommunications provider. The research adopts the Cross-Industry Standard Process for Data Mining (CRISP-DM) as its guiding methodology, encompassing business understanding, data exploration, preprocessing, model development, evaluation, and deployment design. The dataset consists of 8,454 business customers, including small, medium, and large enterprises, described by 14 original variables and a binary churn indicator. Data preparation involved systematic handling of missing values, logarithmic transformation of skewed revenue variables, categorical encoding, feature scaling, and the creation of domain driven behavioral and financial interaction features, resulting in a final feature set of 22 variables. Given the severe class imbalance, with churners accounting for approximately 6.5% of observations, several resampling techniques were assessed. SVMSMOTE demonstrated the most effective improvement in minority-class learning. A diverse set of classifiers was subsequently evaluated, including ensemble-based methods, gradient boosting models, neural networks, and stacked ensembles. Performance was measured using imbalance-sensitive metrics such as recall, F1-score, precision-recall area under the curve, and Matthews correlation coefficient. The empirical results show that ensemble learning approaches provide superior churn detection compared to standalone models. The EasyEnsemble classifier combined with SVMSMOTE achieved the most favorable balance between sensitivity and precision, attaining an F1-score of 0.129 and a recall rate of 38.2%. Business impact analysis indicates that the model successfully identified over one-third of churned customers, demonstrating its potential to support proactive retention strategies despite moderate campaign efficiency. Overall, the study demonstrates that effective churn prediction in enterprise telecommunications environments requires imbalance-aware modeling, carefully engineered features, and evaluation metrics aligned with business risk. The framework offers a scalable and operationally relevant solution that links predictive analytics with actionable customer retention decision-making.

Keywords: Random Forest, Customer Churn, Telecommunication, Customer Retention

1. Introduction

Within the global telecommunications industry, customer churn refers to the discontinuation of subscription-based services by users who migrate to competing providers. This phenomenon represents one of the most persistent challenges

facing telecommunications operators, with direct implications for profitability, operational efficiency, and long-term market competitiveness. High churn rates erode revenue stability and increase operational costs, particularly in mature and highly competitive markets. Recent estimates indicate that annual churn rates in the global telecommunications

ecosystem exceed 30 percent [8], underscoring the urgency for data-driven prediction and customer retention mechanisms. The financial consequences of such attrition are substantial, prompting operators to adopt proactive strategies aimed at identifying and retaining customers before service termination occurs.

Mobile communication continues to underpin global connectivity, with smartphones and feature phones accounting for the majority of voice and data usage worldwide. Although the market has reached a high level of saturation, the mobile segment remains one of the most dynamic areas of the telecommunications sector. As competitive pressure intensifies, particularly in post-COVID market conditions, the strategic emphasis of operators has shifted from aggressive subscriber acquisition toward customer retention. Churn, defined as the proportion of users who discontinue service in favor of alternative telecommunication service providers [8], has therefore become a central performance indicator for telecommunications firms. Evidence suggests that annual customer losses ranging between 30 and 35 percent have been observed in several markets during periods of economic disruption. These trends are further influenced by pricing strategies, where increases by some operators and reductions by others encourage customer migration across networks. At the core of this challenge lies the high cost of customer acquisition, which significantly exceeds the investment required to retain existing subscribers, making churn management a strategic necessity rather than a reactive response.

From an economic perspective, customer retention is consistently shown to be more cost-effective than customer acquisition. Empirical studies indicate that acquiring a new subscriber may cost between five and seven times more than maintaining an existing one [6]. Elevated churn rates destabilize revenue streams and constrain an organization's ability to remain competitive in an increasingly crowded marketplace. The rapid technological advancement of the telecommunications industry has lowered entry barriers, enabling new players to introduce innovative service offerings and intensifying competitive rivalry. In such an environment, churn is driven by market saturation, aggressive promotional strategies, and frequent introduction of bundled services. Telecommunications operators must therefore balance growth objectives with sustainable retention strategies, positioning churn prediction as a core component of modern operational and strategic planning.

Customer Churn Prediction (CCP) has emerged as a critical analytical tool for anticipating customer attrition and supporting targeted retention initiatives. According to [6], effective churn prediction enables organizations to design timely interventions that enhance revenue performance and strengthen competitive positioning. Telecommunications firms generate extensive volumes of customer data, including service usage records, billing histories, demographic profiles, and interaction logs. These datasets offer valuable insights into behavioral patterns associated with churn risk. The principal challenge lies not in data availability, but in the

ability to transform these data into actionable intelligence capable of identifying early indicators of disengagement before customers terminate their subscriptions.

Globally, telecommunications and other service-oriented industries continue to experience persistent churn as digitalization accelerates and customer switching costs decline. While the sector generates significant revenue, particularly in developing economies, operators face increasing difficulty in retaining subscribers amid rapid technological change and evolving consumer expectations. International research has demonstrated the effectiveness of machine learning techniques in churn prediction, employing approaches such as ensemble learning, feature selection, and gradient boosting to improve predictive accuracy. However, emerging literature emphasizes the need to move beyond purely predictive objectives and incorporate explanatory insights that clarify the underlying drivers of churn. Studies such as [9] illustrate that models including Logistic Regression, Random Forests, LightGBM, and ensemble frameworks can successfully identify high-risk customers across diverse datasets. Nevertheless, in regions where telecommunications services play a critical role in financial inclusion, education, and everyday communication, predictive performance must be complemented by interpretability to ensure that retention strategies align with local behavioral and market dynamics.

At the national level, research within Kenya highlights the importance of customer experience management in mitigating churn. The study by [10], based on survey data collected from telecommunications employees, demonstrates that social interactions, quality of service interfaces, and the retail environment contribute positively to customer retention, while pricing pressures increase churn propensity. These findings suggest that customer loyalty is influenced by technical service quality, relational, and experiential factors. As telecommunications services become increasingly embedded in daily life, operators must adopt customer-centered strategies that integrate social engagement, improved service interactions, and affordable pricing structures to sustain market stability.

The rapid pace of digital adoption, combined with socio-economic pressures affecting mobile service consumption, has further intensified competition within the telecommunications market. The integration of demographic attributes, revenue indicators, and service usage patterns provides operators with an opportunity to implement informed retention interventions. By leveraging such data-driven approaches, telecommunications firms can reduce unnecessary marketing expenditure, strengthen customer loyalty, and enhance long-term profitability. Consequently, effective churn prediction is not merely a technical exercise but a strategic instrument for sustaining competitiveness in an increasingly volatile telecommunications landscape.

Research Objectives

The aim of this study is to address existing limitations in customer churn prediction within the telecommunications industry by developing a data-driven and decision-oriented analytical framework. The research seeks to enhance both the predictive accuracy and practical usability of churn prediction systems, ensuring alignment between analytical outputs and managerial decision-making needs.

Main Objective

To design and implement a robust machine learning framework for predicting customer churn in the telecommunications industry.

Specific Objectives

- i. To critically analyze the methodological and conceptual shortcomings identified in existing studies on customer churn prediction within the telecommunications sector.
- ii. To construct and train customer churn prediction models using selected machine learning algorithms and telecommunications customer data.
- iii. To assess the predictive performance of the developed models through the application of established classification evaluation metrics.
- iv. To operationalize the most effective churn prediction model by integrating it into an interactive and interpretable dashboard that supports informed managerial decision-making.

This study contributes to both theoretical and practical understanding of customer churn in telecommunications. By integrating machine learning techniques with business-oriented decision support mechanisms, the research advances a sustainable, data-informed approach to customer retention. The findings are intended to support the translation of advanced analytics into everyday operational practice, strengthening the role of predictive intelligence in enhancing customer loyalty and long-term organizational performance.

2. Literature Review

There has been a substantial transformation in the digital landscape over recent years, characterized by rapid technological advancement and the emergence of new markets and service-oriented startups. These developments have significantly reshaped customer expectations within the telecommunications industry, where demand increasingly favors affordable, reliable, and personalized service offerings. While digital evolution has broadened access to communication and enhanced service availability, it has simultaneously intensified competitive pressures across the sector. In this environment, retaining existing subscribers has become more strategically important than the continual acquisition of new customers.

A central challenge confronting telecommunications operators is the customer's decision to discontinue service, often driven by pricing disparities, service quality perceptions, or competitive incentives. Evidence indicates that several

operators have lost up to one-third of their subscriber base to rival networks, elevating customer churn to a critical global concern. Such losses have directly translated into sustained revenue erosion and weakened market stability. Consequently, there is a growing imperative for the development and adoption of data-driven approaches capable of identifying churn risk early and supporting proactive retention strategies [5].

2.1. Theoretical Framework

2.1.1. Customer Relationship Management

Customer Relationship Management (CRM) has consistently been identified as a core strategic framework for sustaining customer loyalty and long-term organizational performance. [11] emphasized that while organizations continue to invest heavily in customer acquisition, retention has become equally critical as competitive markets intensify. Even marginal increases in churn rates can result in disproportionate revenue losses, underscoring the importance of early detection and intervention. Effective CRM therefore, depends on a data-informed understanding of customer behavior, satisfaction, and cognitive signals that precede disengagement or dissatisfaction.

The authors further noted that the effectiveness of machine learning based churn prediction systems is largely determined by the quality of feature engineering. Traditional CRM-driven churn studies have predominantly relied on structured attributes such as demographic characteristics, service usage patterns, tenure, and billing information. Although these variables provide valuable quantitative indicators, they frequently fail to capture qualitative aspects of customer-provider interaction, including sentiment, tone, and expressed intent embedded in textual communications.

To overcome these limitations, [11] introduced the Customer Churn-related Knowledge Base (ChurnKB), a domain-informed framework designed to enhance feature representation through the incorporation of unstructured customer-generated content, including emails, chat transcripts, and feedback messages. This approach marked a methodological shift from purely numeric representations toward linguistically derived behavioral indicators, thereby extending the predictive capacity of conventional CRM systems.

ChurnKB employs a suite of Natural Language Processing (NLP) techniques, including Term Frequency-Inverse Document Frequency (TF-IDF), cosine similarity, regular expressions, tokenization, and stemming, to extract churn-relevant linguistic patterns. These features enable the identification of early indicators of dissatisfaction, such as complaints, negative sentiment, or cancellation intent, which may not be evident in structured datasets alone. To further enhance adaptability, the framework integrates Generative AI models, particularly Large Language Models (LLMs), allowing the system to detect latent emotional and cognitive cues within unstructured text. Feedback mechanisms embedded in the framework support continuous learning, ensuring alignment with evolving customer behavior.

Empirical evaluation demonstrated that incorporating knowledge-enhanced features led to substantial improvements in predictive performance across several machine learning models, including Logistic Regression, Random Forest, Multilayer Perceptron, and XGBoost. The most pronounced improvement was observed in the XGBoost model, where the F1-score increased from 0.5752 to 0.7891, illustrating the effectiveness of knowledge-driven feature enrichment in identifying at-risk customers and enabling earlier intervention.

Beyond churn prediction, the framework exhibited applicability in related domains such as personalized marketing, sentiment analysis, cyberbullying detection, hate speech recognition, and mental health monitoring, highlighting the broader relevance of integrating CRM principles with AI-driven behavioral analytics. However, [11] cautioned that the integration of LLMs introduces challenges related to data governance, ethical use of AI, and model interpretability. The authors emphasized the need for Explainable AI (XAI) techniques to ensure that model outputs remain transparent and actionable for business stakeholders, as well as the importance of validating such frameworks across diverse sectors and regional contexts.

2.1.2. Predictive Modeling Using Statistical and Segmentation Approaches

The telecommunications industry is particularly vulnerable to customer attrition, as even modest increases in churn can have immediate and significant effects on profitability. [22] emphasized that retaining existing subscribers is substantially more cost-effective than acquiring new ones, reinforcing the importance of predictive modeling for early churn detection. Their study focused on identifying behavioral and demographic characteristics that differentiate loyal customers from those likely to discontinue service.

The research utilized customer data from three major Chinese telecommunications operators, incorporating demographic, usage-based, and service-related variables. A key methodological component was customer segmentation, which enabled the identification of homogeneous groups exhibiting similar churn tendencies. This segmentation facilitated the development of targeted retention strategies tailored to the behavioral profiles of distinct customer groups.

Results indicated that logistic regression achieved superior predictive accuracy compared to Fisher discriminant analysis, attaining an accuracy of 93.94 percent. The findings highlighted the robustness of logistic regression in handling categorical predictors and nonlinear relationships when combined with effective segmentation and variable selection. However, the reliance on structured numerical data and the absence of behavioral or textual variables revealed a methodological gap, suggesting opportunities for hybrid approaches that integrate traditional statistical modeling with machine learning techniques.

2.1.3. Forest Models and Ensemble Strategies

Ensemble learning approaches, particularly decision forest models, have increasingly become central to churn prediction

in telecommunications. [6] argued that single rule-based classifiers, while interpretable, often lack the robustness and scalability required for large, high-dimensional customer datasets. Their study demonstrated that class imbalance, where churners represent a small minority, significantly limits the effectiveness of conventional models such as Logistic Regression or standalone Decision Trees.

To address these challenges, the authors proposed a suite of Decision-Forest architectures, including Logistic Model Trees, Random Forests, and Functional Trees. By aggregating multiple weak learners, these ensembles reduce variance and overfitting while exploiting complementary classifier strengths. The approach proved particularly effective in handling imbalanced datasets, improving recall for churners without disproportionately increasing false positives.

Using benchmark datasets, ensemble models employing weighted stacking and soft-voting outperformed baseline classifiers in both precision and recall. The findings confirmed that forest-based ensembles are well-suited for churn prediction problems characterized by noise, non-linearity, and class imbalance. Despite their strong performance, the authors emphasized the importance of integrating explainability mechanisms, recommending the incorporation of SHAP or LIME to enhance managerial trust and operational adoption.

2.1.4. Machine Learning Approaches for Churn Prediction

[20] examined the application of supervised machine learning techniques for churn prediction, comparing traditional decision tree models with Random Forests and Support Vector Machines. The study highlighted that earlier decision tree-based systems often suffered from redundant feature inclusion, leading to overfitting and limited predictive accuracy.

To address these issues, the authors proposed a hybrid approach combining Random Forests for feature selection with Support Vector Machines for classification. Random Forests reduced feature redundancy through automatic importance ranking, while SVMs enhanced class separation. The hybrid model achieved an accuracy of 95 percent, representing a substantial improvement over traditional decision tree approaches. While effective, the study relied solely on structured data, leaving opportunities for future work incorporating real-time analytics, unstructured data, and explainable modeling techniques.

2.2. Empirical Studies

2.2.1. Handling Class Imbalance in Customer Churn Prediction

[5] investigated the effects of class imbalance on churn prediction using customer data from a major telecommunications operator in Nepal. The dataset comprised over 52,000 records, with churners constituting a minority. XGBoost was employed to address imbalance, achieving 97 percent accuracy and an F1-score of 88 percent on the native dataset.

The study emphasized the importance of evaluating churn models using metrics beyond accuracy, such as recall, F1-score, and ROC-AUC, which better reflect performance on minority classes. While the results demonstrated strong predictive capability, the absence of explainability techniques and comparative resampling strategies highlighted areas for further methodological enhancement.

2.2.2. Hybrid Approaches to Churn Prediction

[21] proposed a hybrid framework combining classification and clustering to improve churn prediction and interpretability. Feature selection using Information Gain and Correlation Ranking reduced redundancy, while Random Forest classification achieved the highest accuracy among the tested models. Subsequent k-means clustering enabled segmentation of churned customers into behaviorally similar groups, supporting targeted retention interventions.

Despite improved performance and insight generation, the framework lacked explainable AI components and did not account for temporal dynamics, indicating opportunities for future research incorporating time-aware and explainable modeling approaches.

2.2.3. Model Optimization and Deep Learning

[7] evaluated traditional machine learning, ensemble methods, and deep learning architectures across international telecommunications datasets. Deep learning models, particularly CNNs and ANNs, achieved near-perfect accuracy; however, concerns regarding interpretability and class imbalance remained. The authors concluded that while deep models capture complex nonlinear relationships effectively, their practical adoption requires transparency and business-aligned evaluation.

2.2.4. Knowledge-Based and Text-Driven Feature Engineering

Recent advances by [11] further demonstrated the value of knowledge-based and text-driven feature engineering in churn prediction. By integrating linguistic and cognitive indicators extracted from unstructured text into machine learning models, the study achieved substantial improvements in predictive performance. The findings reinforce a growing shift toward models that balance accuracy with interpretability and contextual understanding.

2.2.5. Customer Churn in the Telecommunications Industry

Customer churn remains a critical strategic and financial challenge in telecommunications. Global churn rates exceed 30 percent annually [8], driven by intense competition and low switching costs. Studies integrating ensemble learning with Explainable AI demonstrate that predictive accuracy can be achieved alongside interpretability, enabling actionable insight into churn drivers such as contract type, usage behavior, and payment history.

Complementary research in Kenya [10] highlights the role of customer experience, service interfaces, and pricing

strategies in retention, reinforcing the behavioral dimension of churn management. Together, these findings demonstrate that effective churn mitigation requires both predictive analytics and experience-driven intervention strategies.

2.2.6. Industry Relevance and Contribution

Advancements in predictive modeling have transformed churn management from descriptive analysis to proactive decision support. Ensemble and deep learning models offer superior accuracy, while Explainable AI ensures transparency and managerial trust. This study contributes by integrating these dimensions into a scalable, interpretable, and context-aware framework that bridges predictive modeling with operational decision-making.

2.3. Conceptual Framework

The conceptual framework adopted in this study integrates behavioral profiling, retention insights, and model evaluation as interdependent components of churn prediction. By aligning machine learning outputs with CRM-driven insights, the framework ensures technical rigor while supporting managerial relevance.

Behavioral and Demographic Profiling

Behavioral and demographic attributes such as tenure, revenue, usage frequency, and inactivity patterns provide early indicators of churn risk. These variables form the analytical foundation for model development in this study.

Retention Insights

Predictive outputs are translated into actionable retention strategies through CRM integration, supporting proactive intervention and long-term customer loyalty.

2.4. Gaps in Existing Literature

Despite extensive research, churn prediction literature remains constrained by limited feature diversity, narrow evaluation metrics, and insufficient interpretability. Addressing these gaps requires models that integrate financial relevance, behavioral depth, and explainability.

2.4.1. Limited Feature Diversity

A recurring limitation across the literature is the overreliance on structured Customer Relationship Management (CRM) attributes, such as usage duration, billing frequency, and tenure, as primary churn predictors. [10] observed that while transactional and behavioral variables are useful, they often fail to incorporate deeper financial metrics such as Average Revenue per User (ARPU), Total Revenue, or service profitability indices, which are crucial for business forecasting and segmentation. Similarly, [5] and [7] emphasized that the absence of diversified features like sentiment indicators, customer engagement scores, or network experience measures can reduce model robustness and limit its interpretive value. Expanding feature representation to

include financial, psychological, and contextual dimensions is therefore, necessary for building models that align with both customer behavior and firm-level profitability.

2.4.2. *Narrow Evaluation Practices*

A second gap in churn prediction research lies in the evaluation of model performance. Most studies emphasize metrics such as accuracy, recall, or AUC, which are effective for assessing statistical performance but insufficient for understanding business implications. [11] highlighted that accuracy-driven evaluation frameworks often fail to account for the financial cost of misclassifications, particularly false negatives (missed churners), which can translate to significant revenue losses. Moreover, very few studies integrate return-on-investment (ROI) or customer lifetime value (CLV) indicators into churn evaluation, despite their direct connection to managerial decision-making. Integrating financial metrics with predictive performance would enable a more comprehensive understanding of model impact, bridging the gap between data science outputs and strategic business outcomes.

2.4.3. *Interpretability Limitations*

Despite impressive advances in model accuracy through ensemble and deep learning methods, interpretability remains a major obstacle to operational adoption. Complex ML models such as Random Forest, Gradient Boosting, and Convolutional Neural Networks often operate as opaque, black-box systems, which makes it difficult for managers to understand how predictions are generated [8]. The lack of transparency in feature importance and decision logic reduces managerial trust and delays the integration of predictive models into everyday CRM operations. Recent efforts in Explainable Artificial Intelligence (XAI), including tools like SHAP and LIME, have improved post-hoc interpretability; however, their use in telecommunication churn literature remains limited [6]. Therefore, a critical research need is to embed interpretability mechanisms into model design rather than as an afterthought, ensuring that predictive systems are both accurate and comprehensible.

This study addresses these shortcomings by expanding feature representation, incorporating ROI-linked evaluation, and prioritizing interpretable machine learning frameworks.

3. Methodology

This chapter describes the methodological framework adopted to design, develop, evaluate, and operationalize a customer churn prediction system within the telecommunications sector. The methodology emphasizes analytical rigor, reproducibility, and business relevance by systematically integrating statistical learning and machine learning techniques with domain-specific and decision-making requirements.

This study is structured using the Cross-Industry Standard Process for Data Mining (CRISP-DM), a widely adopted

framework for applied data science projects. This provides a clear and iterative structure consisting of six interconnected phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment. Adopting this framework ensures that each analytical decision is informed by business context, particularly the financial and operational consequences of customer churn in competitive telecommunications markets.

The empirical analysis is based on real operational data obtained from a leading Bulgarian telecommunications provider and focuses exclusively on business-to-business customers. By combining predictive modeling with exploratory and explanatory analysis, the methodology aims to identify customers at risk of churn and to generate interpretable insights that support proactive customer retention strategies.

3.1. *Business Understanding*

In the telecommunications industry, customer churn represents a critical threat to revenue stability and long-term competitiveness, especially within business customer segments where account values are high and service portfolios are complex. As market saturation increases and price-based competition intensifies, retaining existing customers becomes significantly more cost-effective than acquiring new ones. Consequently, the ability to anticipate customer defection before it occurs is a strategic necessity.

From a business perspective, the primary objective of this study is to develop a data-driven system capable of identifying business customers with a high likelihood of churn. Such a system enables targeted retention interventions, efficient allocation of marketing resources, and improved customer lifetime value management. This objective directly informs subsequent analytical choices, including feature selection, model evaluation criteria, and deployment considerations.

3.2. *Data Understanding*

The dataset analyzed in this study consists of real-world records extracted from the business information system of a major Bulgarian telecommunications operator. The data focus on business customers, including small, medium, and large enterprises, and reflect actual service usage, subscription activity, and revenue generation behavior.

In total, the dataset contains 8,454 customer observations described by 14 explanatory variables and one binary target variable indicating churn status. The variables capture multiple dimensions of customer behavior and value, including customer segmentation, account management, subscriber activity, and financial performance. Operational attributes such as the number of active, inactive, and suspended subscribers provide insight into engagement and service continuity, while financial indicators including average mobile revenue, average fixed revenue, total revenue, and average revenue per user (ARPU) represent the economic contribution of each customer.

Variable	Description
PID	Personal identification number of the business customer
GCRM PID Value Segment	Customer value classification in the telecom system
Billing ZIP	Location ZIP code of the business customer
KA Name	Assigned Key Account Manager
Active Subscribers	Number of active subscribers
Not Active Subscribers	Number of inactive subscribers
Suspended Subscribers	Subscribers suspended in the last six months
Total Subscribers	Total subscribers under the customer
Average Mobile Revenue	Mean mobile service revenue per subscriber
Average Fixed Revenue	Mean fixed service revenue per subscriber
Total Revenue	Total customer revenue
ARPU	Average revenue per user
Churn	Churn status (Yes/No)

Table 1. Description of Variables Used in the Customer Churn Dataset

3.3. Exploratory Data Analysis

Preliminary exploration revealed a substantial class imbalance, with approximately 6.49% of customers classified as churned. This imbalance is typical of telecommunications datasets and has important implications for model selection and evaluation, as standard classifiers may otherwise favor the majority non-churn class. Initial inspection also identified missing values across several variables, necessitating a structured data preparation strategy.

Further exploratory analysis indicates that churn behavior is not uniformly distributed across customer segments. Differences are observed among effective customer categories such as SOHO, SME, and VSE, suggesting that segmentation characteristics influence churn propensity and should be incorporated into predictive modeling.

Revenue-related patterns also show meaningful variation between churned and retained customers. Analysis of total revenue distributions suggests that churn affects both low- and high-value customers, highlighting the financial risk associated with customer attrition and reinforcing the importance of revenue-based predictors in churn modeling.

The dataset provides a realistic and comprehensive representation of business customer dynamics within the telecommunications sector. Its combination of operational, financial, and segmentation attributes makes it well-suited for developing and validating churn prediction models that are both analytically robust and directly applicable to managerial decision-making.

3.4. Data Cleaning and Preprocessing

This constituted a critical phase in preparing the dataset for reliable churn prediction. Given that the data were extracted from an operational telecommunications business information system, the dataset exhibited characteristics common to real-world enterprise data, including missing values, scale heterogeneity, and skewed financial variables. A structured and systematic preprocessing pipeline was therefore implemented to enhance data quality, ensure numerical stability, and improve model performance.

3.4.1. Handling Missing Values

Missing operational values in subscriber-related fields, such as the number of non-active and suspended subscribers, were replaced with zeros. This choice reflects the operational assumption that missing records in these fields indicate the absence of such events rather than unreported activity.

Categorical attributes with missing entries, including the CRM customer value segment, were assigned an explicit *Unknown* category. This approach preserves potentially informative missingness while avoiding data loss.

For numerical location and revenue-related variables, median-based imputation was applied. Specifically, missing values in billing ZIP codes and Average Revenue Per User (ARPU) were replaced using median statistics, which are robust to extreme values and skewed distributions frequently observed in telecommunications revenue data.

3.5. Feature Engineering

Following data cleaning, an advanced feature engineering strategy was implemented to enhance the predictive capacity of the churn models. The objective of this stage was to transform raw operational and financial variables into informative representations that capture customer engagement, revenue efficiency, and risk exposure. All engineered features were derived using domain knowledge from telecommunications operations and were designed to improve model sensitivity to early churn signals.

3.5.1. Categorical and Numerical Feature Specification

The final feature space comprised 22 variables, grouped into categorical and numerical attributes. Categorical features included customer value segmentation, effective business segment, and key account manager identifiers. These attributes capture structural differences in customer type, contractual relationships, and account servicing, all of which may influence churn behavior.

Numerical features encompassed subscriber activity metrics, revenue indicators, and engineered behavioral ratios. Variables such as the number of active, inactive, and suspended subscribers reflect service engagement and continuity. Financial attributes including total revenue, average mobile revenue, average fixed revenue, and ARPU capture the economic contribution of each business customer.

3.5.2. Derived Behavioral and Financial Indicators

To enrich the explanatory power of the dataset, several composite features were constructed. Revenue ratios were introduced to reflect the dependence of customers on specific service types relative to total revenue. Engagement efficiency was captured through activity rates, defined as the proportion of active subscribers relative to the total subscriber base.

A risk score was computed by combining inactive and suspended subscriber counts, normalized by total subscriptions. This indicator serves as a proxy for service instability and disengagement, which are known precursors to churn. Interaction features were also generated to capture joint

effects between revenue magnitude and customer engagement, enabling models to learn non-linear dependencies between financial value and behavioral risk.

3.5.3. Transformation of Skewed Variables

Several revenue-related variables exhibited right-skewed distributions, a common characteristic of enterprise customer data. To mitigate the influence of extreme values and improve numerical stability, logarithmic transformations were applied to total revenue, ARPU, average mobile revenue, and average fixed revenue. These transformations preserved relative differences while reducing variance and improving compatibility with both linear and non-linear classifiers.

3.5.4. Feature Encoding and Scaling

Categorical features were transformed using label encoding to convert non-numeric values into machine-readable representations. Encoders were fitted exclusively on training data and applied consistently to the test set to prevent information leakage.

All numerical features were standardized using z-score normalization. This step ensured that variables with larger numeric ranges did not dominate model learning, particularly for distance-based, gradient-based, and neural network models. Scaling parameters were derived from the training set only and subsequently applied to the test data to maintain evaluation integrity.

After preprocessing, the final modeling dataset consisted of 22 features and was divided into training and testing subsets using an 80:20 stratified split. The training set contained 6,762 observations, while the test set comprised 1,691 observations. Stratification preserved the original churn distribution across both subsets, ensuring unbiased model evaluation.

3.6. Modeling

Customer churn prediction was treated as a supervised binary classification problem, with the objective of distinguishing churned customers from retained ones. Given the strong class imbalance present in the dataset, the modeling strategy emphasized recall, balanced accuracy, and probabilistic discrimination rather than overall accuracy alone.

To achieve robust and generalizable results, a comprehensive modeling pipeline was implemented, integrating advanced sampling techniques, ensemble learning methods, gradient boosting frameworks, and neural networks.

3.6.1. Class Imbalance Handling

Multiple resampling strategies were evaluated to address the underrepresentation of churned customers in the training data. These included SMOTE, ADASYN, Borderline-SMOTE, SVM-SMOTE, and SMOTE combined with Tomek links. Each technique generates synthetic minority class samples using different neighborhood and boundary-based strategies, thereby improving model sensitivity to churn cases.

Sampling was applied exclusively to the training data to avoid introducing bias into the test set. This ensured that

performance metrics reflected real-world predictive behavior.

3.6.2. Advanced Machine Learning Models

A diverse set of state-of-the-art classifiers was trained to capture varying levels of model complexity and interpretability. Ensemble-based methods included EasyEnsemble, RUSBoost, and Balanced Bagging classifiers, which combine resampling and ensemble learning to improve minority class detection.

Gradient boosting models were implemented using XGBoost, LightGBM, and Histogram-Based Gradient Boosting. These models are well suited for high-dimensional, non-linear data and incorporate mechanisms to handle class imbalance through weighting and loss optimization. Hyperparameters such as tree depth, learning rate, and number of estimators were selected to balance predictive accuracy and generalization.

Where available, CatBoost was included due to its robustness to categorical variables and ability to mitigate overfitting through ordered boosting.

3.6.3. Neural Network Architecture

In addition to tree-based models, a deep learning classifier was developed using a multilayer perceptron architecture. The network comprised multiple dense layers with rectified linear unit activation functions, batch normalization, and dropout regularization. These mechanisms reduce overfitting and improve convergence stability.

To address class imbalance, class weights were incorporated directly into the loss function. Early stopping based on validation loss was applied to prevent overtraining and to restore optimal model weights. The network output represented churn probability through a sigmoid activation function.

3.6.4. Ensemble Learning Strategies

To further enhance predictive robustness, ensemble meta-models were constructed using both soft voting and stacking strategies. Voting classifiers aggregated predictions from gradient boosting and balanced random forest models using probabilistic averaging. Stacking classifiers combined base learner outputs with a gradient boosting model as a meta-classifier, allowing higher-level learning from model complementarities. These ensemble strategies reduce variance, improve generalization, and leverage the strengths of heterogeneous classifiers.

The advanced modeling pipeline enabled systematic comparison across sampling techniques, individual classifiers, deep learning models, and ensemble strategies. By integrating feature engineering, imbalance handling, and diverse learning paradigms, the modeling framework was designed to identify high-risk churn customers accurately while maintaining robustness and scalability for operational deployment.

3.7. Model Evaluation

Model performance was assessed using a comprehensive set of metrics, including accuracy, balanced accuracy, precision, recall, F1-score, ROC-AUC, precision-recall AUC, and Matthews correlation coefficient. Given the business cost associated with failing to identify churners, recall and precision-recall AUC were prioritized during model comparison.

All evaluation metrics were computed on the untouched test dataset to ensure objective assessment of generalization performance. The final model selection balanced predictive effectiveness, robustness under class imbalance, and suitability for operational deployment.

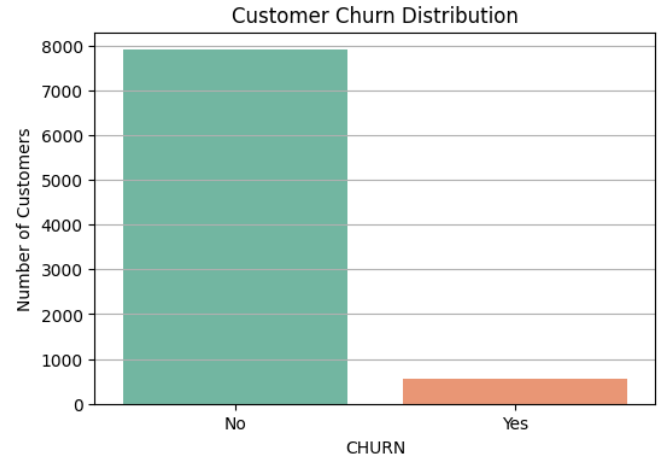


Figure 1. Overall distribution of churned and non-churned business customers

3.8. Model Deployment

The selected model architecture supports integration into an API-driven environment, enabling churn scoring and periodic batch risk assessment. This design ensures scalability, retraining capability, and alignment with customer relationship management systems.

4. Results

This presents the empirical findings obtained from the advanced customer churn prediction framework developed in this study. All performance metrics are computed on an unseen test dataset to ensure unbiased evaluation of predictive behavior under real-world conditions.

4.1. Class Imbalance Assessment

An initial assessment of the target variable confirmed a pronounced imbalance between churned and retained business customers. Only 6.49% of customers were observed to have churned, while the remaining majority continued their subscriptions. This imbalance has direct implications for model training and evaluation, particularly in the selection of appropriate performance metrics.

Figure 1 visually confirms the dominance of the non-churn class, reinforcing the need for imbalance-aware sampling and recall-oriented evaluation in subsequent modeling stages.

4.1.1. Evaluation of Sampling Strategies

To mitigate the effects of class imbalance, multiple oversampling and hybrid resampling strategies were evaluated using a consistent baseline gradient boosting classifier. Performance comparison focused on F1-score and precision-recall area under the curve (PR-AUC), as these metrics emphasize minority class detection.

SVMSMOTE achieved the highest F1-score among all evaluated strategies, outperforming standard oversampling approaches such as SMOTE and ADASYN. Based on this empirical evidence, SVMSMOTE was selected as the optimal resampling method and applied during advanced model training.

4.2. Comparative Performance of Advanced Models

A diverse set of ensemble models, gradient boosting classifiers, and neural network architectures was trained using the selected resampling strategy. Model performance varied considerably across evaluation metrics, highlighting trade-offs between recall, precision, and overall discrimination ability.

4.2.1. Model Ranking by F1-Score

Models were first ranked according to F1-score to assess balanced performance between precision and recall. The EasyEnsemble classifier achieved the highest F1-score, followed by CatBoost and RUSBoost.

Figure 2 shows that ensemble-based classifiers dominate the upper performance range, while several high-accuracy models exhibit comparatively low F1-scores.

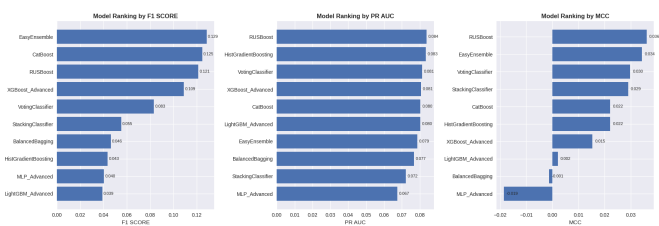


Figure 2. Ranking of advanced models based on F1-score

4.2.2. Precision-Recall Performance

Precision-recall analysis was conducted to evaluate model effectiveness under severe class imbalance. RUSBoost achieved the highest PR-AUC, indicating superior capability in identifying churners across varying decision thresholds. Precision-recall curves were plotted for all advanced models to visualize performance stability across decision thresholds. These curves provide insight into how model precision degrades as recall increases, a critical consideration in churn prevention campaigns. The results demonstrate that ensemble and boosting-based models consistently outperform individual learners in precision-recall space.

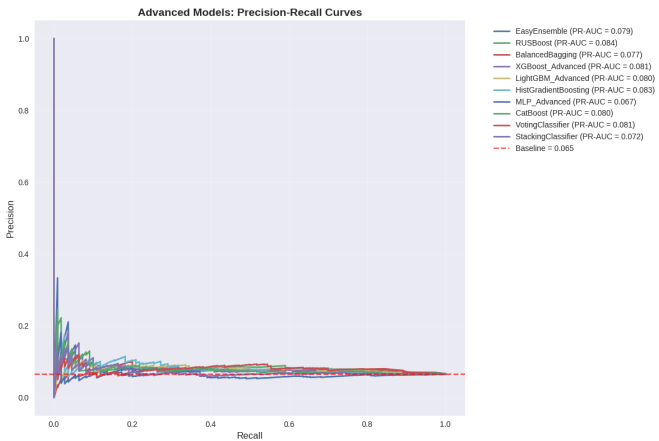


Figure 3. Model ranking based on precision-recall AUC

4.2.3. Integrated Metric Comparison

To facilitate holistic comparison, model performance across all evaluation metrics was summarized using a heatmap representation. Figure 4 highlights the trade-offs between recall-driven models and precision-oriented classifiers, confirming the absence of a universally dominant model.

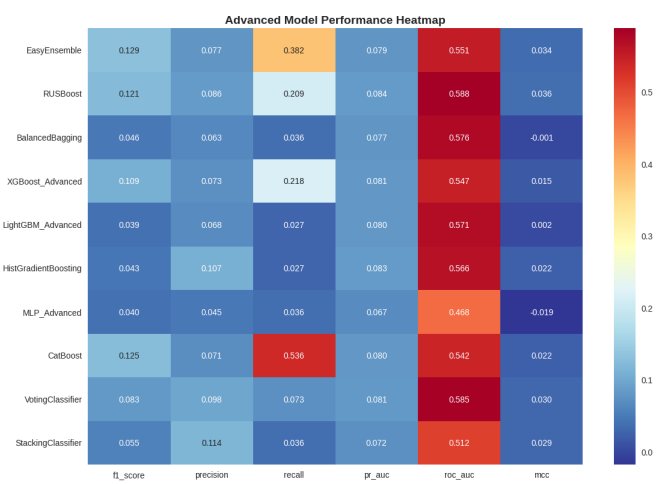


Figure 4. Heatmap of advanced model performance across evaluation metrics

4.3. Selection of the Optimal Sampling Strategy

Following the identification of class imbalance within the dataset, multiple resampling techniques were evaluated to improve minority class representation. The evaluation was conducted using a consistent baseline gradient boosting classifier, and performance was assessed using F1-score and precision-recall area under the curve (PR-AUC). These metrics were selected due to their sensitivity to minority class detection and their relevance in churn prediction contexts. Table 2 shows that SVMSMOTE achieved the highest F1-score among all evaluated resampling approaches. This indicates superior ability to enhance churn detection by generating synthetic samples near complex class boundaries. Based on this empirical evidence, SVMSMOTE was selected as the optimal sampling strategy and applied during all subsequent model training stages.

Sampling Strategy	F1-Score
SMOTE	0.0315
ADASYN	0.0315
SVMSMOTE	0.0654
BorderlineSMOTE	0.0280
SMOTETomek	0.0444

Table 2. Performance comparison of sampling strategies

4.4. Advanced Model Performance Evaluation

Using the selected SVMSMOTE sampling strategy, a diverse set of advanced machine learning and ensemble classifiers was trained and evaluated. All performance metrics were computed on the untouched test dataset to ensure unbiased assessment.

Model	Acc.	Bal. Acc.	Prec.	Recall	F1	ROC	PR	MCC
EasyEnsemble	0.664	0.533	0.077	0.382	0.129	0.551	0.079	0.034
RUSBoost	0.803	0.527	0.086	0.209	0.121	0.588	0.084	0.036
BalancedBagging	0.902	0.500	0.064	0.036	0.046	0.576	0.077	-0.001
XGBoost (Adv.)	0.768	0.512	0.073	0.218	0.109	0.547	0.081	0.015
LightGBM (Adv.)	0.913	0.501	0.068	0.027	0.039	0.571	0.080	0.002
HistGradientBoosting	0.922	0.506	0.107	0.027	0.044	0.566	0.083	0.022
MLP (Adv.)	0.888	0.492	0.046	0.036	0.040	0.468	0.067	-0.019
CatBoost	0.510	0.523	0.071	0.536	0.125	0.542	0.080	0.022
VotingClassifier	0.896	0.513	0.098	0.073	0.083	0.585	0.081	0.030
StackingClassifier	0.919	0.508	0.114	0.036	0.055	0.512	0.072	0.029

Table 3. Performance metrics of advanced churn prediction models

Based on combined F1-score and PR-AUC rankings, the EasyEnsemble classifier was selected as the overall best-performing model. The model achieved a recall of 38.18%, correctly identifying a substantial proportion of churned customers in the test dataset.

4.5. Operational Performance

From an operational standpoint, the selected model correctly identified 42 churned customers, while missing 68. Although the number of false positives was relatively high, this behavior is consistent with churn prevention objectives, where the cost of missing a true churning exceeds the cost of an unnecessary retention action.

Across all evaluated models, ensemble-based approaches demonstrated superior performance in detecting churned customers. EasyEnsemble achieved the highest F1-score, while RUSBoost attained the strongest PR-AUC and Matthews Correlation Coefficient. High-accuracy models such as LightGBM and Histogram Gradient Boosting exhibited limited churn sensitivity, reflecting conservative prediction behavior under class imbalance.

5. Discussion

This focuses on the effectiveness of imbalance handling strategies, comparative model behavior, business relevance of predictive performance, and the broader methodological contributions of the study. Rather than reiterating numerical results, it explains why specific patterns emerged and how they inform both theory and practice.

5.1. Impact of Class Imbalance on Churn Modeling

The pronounced imbalance between churned and retained customers strongly influenced model performance across all experiments. Despite several models achieving high overall accuracy, many failed to identify churners effectively, as evidenced by low recall and near-zero Matthews Correlation Coefficients. This confirms that churn prediction in business telecommunications contexts cannot be treated as a

conventional classification task and instead requires explicit imbalance-aware design.

The findings reinforce the inadequacy of accuracy as a primary evaluation metric in churn analytics. Models optimized for accuracy systematically favored the majority class, masking poor churn detection capability. This observation aligns with the study's decision to prioritize recall, F1-score, and precision-recall AUC as more appropriate indicators of business value.

5.2. Effectiveness of Sampling Strategies

Among the evaluated resampling techniques, SVMSMOTE emerged as the most effective method for improving churn detection. Its superior performance can be attributed to its boundary-focused sample generation, which concentrates observations near complex class margins rather than uniformly oversampling minority instances.

This result suggests that churn behavior in business customers is not randomly distributed but instead occurs within nuanced regions of the feature space where customer engagement and revenue signals overlap. By enhancing representation in these regions, SVMSMOTE enabled downstream models to learn more discriminative decision boundaries. The findings highlight that sampling strategy selection plays a role as critical as model choice in imbalanced churn prediction problems.

5.3. Interpretation of Model Performance Differences

The comparative evaluation revealed that ensemble-based classifiers consistently outperformed individual learners and deep neural networks in identifying churned customers. EasyEnsemble achieved the highest F1-score, reflecting a balanced trade-off between precision and recall, while RUSBoost demonstrated superior ranking ability as indicated by PR-AUC and MCC.

These outcomes indicate that combining resampling with ensemble learning is particularly effective for churn prediction under severe imbalance. By training multiple classifiers on differently balanced subsets, ensemble methods reduce variance and improve minority class sensitivity. In contrast, several gradient boosting and neural network models exhibited conservative prediction behavior, leading to high accuracy but low recall.

The underperformance of the deep learning model highlights an important practical insight. In highly imbalanced, tabular business datasets with limited churn signal strength, complex neural architectures may not provide meaningful advantages over ensemble tree-based methods. This finding challenges the assumption that higher model complexity necessarily yields superior predictive performance in churn analytics.

5.3.1. Trade-offs Between Recall and Precision

The selected best-performing model demonstrated relatively high recall at the expense of precision. From a business

perspective, this trade-off is acceptable and often desirable in churn prevention scenarios. Missing a true churner results in irreversible revenue loss, whereas contacting a non-churning customer typically incurs only marginal campaign costs.

The observed false positive rate therefore does not undermine the operational usefulness of the model. Instead, it highlights the need for predictive outputs to be integrated with customer value segmentation and retention prioritization strategies. When combined with revenue thresholds or CRM-driven targeting rules, the model can support efficient allocation of retention resources.

5.3.2. Business Impact and Practical Relevance

The business impact analysis confirms that the predictive framework delivers actionable value. By correctly identifying more than one-third of churned customers, the model provides telecommunications operators with a meaningful early warning mechanism. Although the campaign efficiency metric remains modest, the churn prevention rate demonstrates the feasibility of proactive intervention.

These results suggest that predictive churn systems should be deployed as decision-support tools rather than automated decision-makers. Human oversight, domain knowledge, and strategic filters remain essential to translating model outputs into effective retention actions.

5.3.3. Methodological Contributions

This study contributes methodologically by demonstrating the combined importance of feature engineering, imbalance handling, and ensemble learning in business-to-business churn prediction. The integration of behavioral ratios, interaction features, and boundary-aware resampling proved essential for uncovering churn patterns in real-world telecommunications data.

The findings also reinforce the value of multi-metric evaluation. Relying on a single performance measure would have led to suboptimal model selection and misleading conclusions. By jointly considering F1-score, PR-AUC, and MCC, the study provides a more comprehensive and realistic assessment of model behavior.

This highlights four central insights. First, class imbalance fundamentally shapes churn prediction outcomes and must be explicitly addressed. Second, boundary-focused resampling techniques outperform traditional oversampling methods in complex churn datasets. Third, ensemble-based models offer the most reliable balance between predictive power and robustness. Finally, predictive accuracy alone is insufficient; business relevance and interpretability are essential for real-world deployment.

6. Conclusion

This study was undertaken to address methodological, analytical, and practical gaps in customer churn prediction within the telecommunications industry, with particular emphasis on business-to-business customers. Guided by

the stated research objectives and questions, the study demonstrates how machine learning can be systematically applied to improve churn detection while maintaining operational relevance for decision-makers.

The first research objective sought to examine methodological and conceptual limitations in existing churn prediction studies. The findings confirm that much of the existing literature emphasizes overall accuracy while neglecting the effects of class imbalance and the financial consequences of missed churners. This limitation was directly observed in baseline and advanced models that achieved high accuracy yet failed to meaningfully detect churned customers. In addressing the first research question, the study establishes that churn prediction models must be evaluated using imbalance-sensitive metrics such as recall, F1-score, and precision-recall AUC rather than accuracy alone, particularly in enterprise telecommunications datasets where churners form a small minority.

The second objective focused on developing churn prediction models using machine learning techniques and telecommunications customer data. This objective was achieved through the implementation of a comprehensive modeling pipeline incorporating feature engineering, imbalance handling, ensemble learning, gradient boosting, and neural networks. In response to the second research question, the results demonstrate that both behavioral indicators (such as subscriber activity and suspension rates) and financial variables (including revenue, ARPU, and derived interaction terms) play a significant role in churn prediction. The integration of engineered features capturing engagement intensity and revenue efficiency substantially improved model sensitivity to churn behavior.

The third objective aimed to evaluate the predictive performance of the developed models using standard classification metrics. Comparative evaluation across multiple classifiers and resampling strategies showed that no single algorithm universally dominates across all metrics. However, in direct response to the third and fourth research questions, the EasyEnsemble classifier combined with SVMOTE consistently achieved the best trade-off between recall and precision, as reflected in the highest F1-score and competitive precision-recall performance. This confirms that ensemble-based imbalance-aware methods are better suited for churn prediction in highly skewed business telecommunications datasets than standalone deep learning or boosting models.

The fourth objective concerned the practical deployment of the optimal model within an interpretable decision-support framework. The study demonstrated how churn predictions can be operationalized through an API-driven architecture and integrated into customer risk dashboards. In addressing the fifth research question, the study illustrates how churn probabilities can be translated into tiered customer risk categories that support strategic retention planning. This linkage between predictive outputs and business actions ensures that model results are statistically valid and actionable within real operational environments.

In conclusion, this research successfully meets its

objectives by delivering a robust, scalable, and imbalance-aware churn prediction framework tailored to business telecommunications customers. The findings confirm that effective churn management requires a combination of advanced machine learning techniques, careful metric selection, and alignment with business decision-making processes. By explicitly connecting predictive performance with financial and operational outcomes, the study contributes to both academic understanding and practical implementation of customer churn analytics. Future research may extend this work by incorporating temporal dynamics, unstructured customer interaction data, and explainable AI techniques to further enhance transparency and predictive reliability.

Abbreviations

ANNs-	Artificial Neural Networks
API-	Application Programming Interface
AUC-	Area Under the Curve
CNNs-	Convolutional Neural Networks
CRISP-DM-	Cross-Industry Standard Process for Data Mining
CRM-	Customer Relationship Management
DL-	Deep Learning
DT-	Decision Tree
ERT-	Extra Random Trees
FT-	Functional Trees
GB-	Gradient Boosting
k-NN-	k-Nearest Neighbors
LightGBM-	Light Gradient Boosting Machine
Lime-	Local Interpretable Model-agnostic Explanations
LMT-	Logistic Model Tree
LR-	Logistic Regression
ML-	Machine Learning
NB-	Naive Bayes
RF-	Random Forest
ROC-	Receiver Operating Characteristic
ROI-	Return on Investment
SHAP-	SHapley Additive exPlanations
SMOTE-	Synthetic Minority Over-sampling Technique
SVM-	Support Vector Machine

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Conflicts of Interest

The authors declare no conflicts of interest.

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